Xiao Long | 隆笑

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OVERVIEW

As an Electronic Information Science and Technology student at Shaanxi Normal University, I am deeply interested in circuit design, power ultrasonics, and acoustic cavitation. During my undergraduate study, I have gained hands-on experience in PCB design, closed-loop control algorithms, C programming, and COMSOL simulation. This entails leading a provincial innovation project, presenting a first-author paper at ICIRA 2026 and securing several awards. Although I am a sophomore without a broad tech stack, I am always eager to learn and pursue interdisciplinary exchange across diverse fields.

EDUCATION

Shaanxi Normal University ***Sep. 2025 – Present***

Electronic Information Science and Technology

3+1 joint undergraduate program with the University of Szeged (Years 1–3 in Xi'an; final year planned in Szeged)

- Average Score: 91.05/100; Academic Rank: 5/113 (Top 4.42%)
- Comprehensive Assessment Rank: 1/113 (Top 0.88%)
- Selected Coursework: Circuit Analysis (96/100); Probability and Mathematical Statistics (96/100); Calculus I & II (95/100, 92/100)

RESEARCH EXPERIENCE

Megasonic Produce Cleaning: System Design and the Cleaning–Integrity Trade-off

Supervisors: Prof. Chenghui Wang; Prof. Qing'an Zhang ***Jan. 2026 – Jul. 2026***

Outline:

- Developed a megasonic produce-cleaning system to study pesticide removal and cellular integrity.

Key Responsibilities:

- Designed control algorithms for automatic frequency sweeping and closed-loop resonance tracking using voltage–current phase feedback.
- Designed controlled experiments varying acoustic frequency, treatment duration, and produce type; combined pesticide-residue screening with optical microscopy of blueberry epidermal tissue to evaluate cleaning efficacy and cellular integrity.

Achievements:

- Identified a cleaning–integrity trade-off: pesticide-residue reduction was accompanied by epidermal cell damage under the tested conditions.
- Authored the paper as first author and presented it in English at ICIRA 2026, Budapest, Hungary (Aug. 2026).

Focused Megasonic Cleaning of Deep Microholes in Semiconductor Chips

Supervisors: Prof. Chenghui Wang; Assoc. Prof. Yifei Sun ***Apr. 2026 – Present***

Outline:

- Leading the development of a focused megasonic cleaning system for deep microholes in semiconductor chips.

Key Responsibilities:

- Designing drive and control circuit boards for the megasonic generator; planning frequency-sweep and resonance-tracking algorithms.
- Simulating focused acoustic fields in COMSOL Multiphysics to evaluate focal position and sound-pressure distribution.

Achievement:

- Project Lead for a Shaanxi provincial-level project under the College Students' Innovation and Entrepreneurship Training Program; recommended for national-level selection.

PUBLICATIONS

[1] Long, X., Xu, H., Wang, C., & Sun, Y. (2026). **Megasonic Based Pesticide Removal from Fresh Produce: System Design and the Cavitation-Induced Cleaning-Integrity Trade-off.** *ICIRA 2026, Budapest, Hungary.* Oral presentation; publication pending.

[2] Feng, Y., Long, X., Zheng, H., Zhang, W., Sun, H., & Wang, X. (2026). **Solving Schrödinger Equation Numerically Using the Matrix Finite Difference Method.** *Physics and Engineering, 46(4), 185–192.* doi:10.27024/j.wlygc.2026.02.26.01.

AWARDS & HONOURS

First-Class Scholarship, Shaanxi Normal University *2026*

Grand Prize (1st Place), 3rd "Maker Cup" Innovation and Entrepreneurship Competition, Shaanxi Normal University; Project Lead, ultrasonic produce-cleaning device project. *2025*

Mixed Doubles Quarterfinalist, School of Physics and Information Technology Badminton Tournament, Shaanxi Normal University. *2025*

SKILLS

English: Fluent; English-medium coursework; IELTS 7.0 overall (Listening 7.5, Reading 9.0, Writing 6.5, Speaking 5.5).

Techniques: AI-assisted coursework and research; C programming; PCB design; COMSOL Multiphysics simulation.